SME206
Signals and Systems

2. Linear Time-Invariant Systems

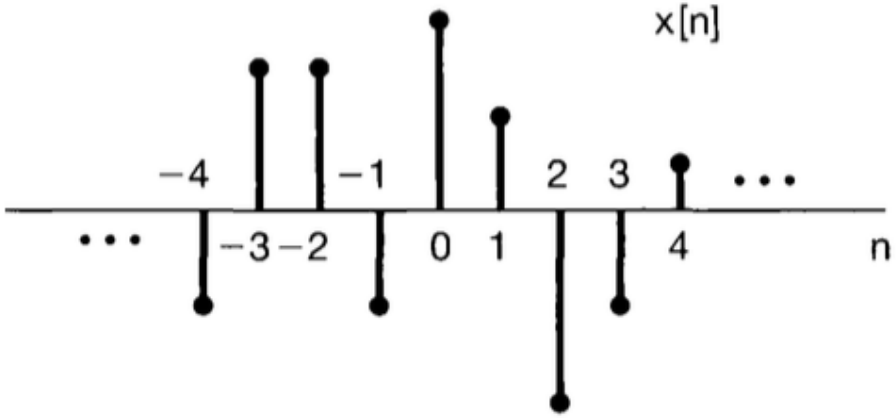
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Discrete LTI Systems

- **Representing the DT signals in terms of impulses**



$$x[n]$$
$$= \cdots + x[-4]\delta[n+4] + x[-3]\delta[n+3] + x[-2]\delta[n+2] + x[-1]\delta[n+1]$$
$$+ x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3] + x[4]\delta[n-4] + \cdots$$

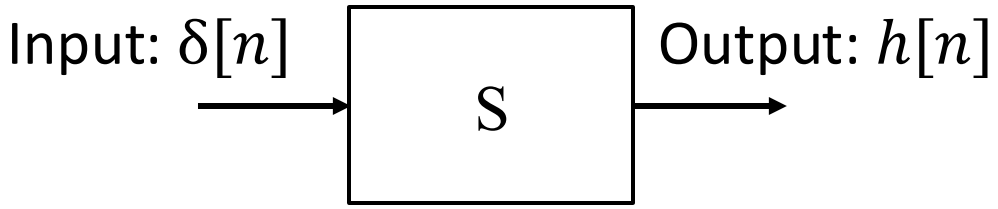
Sifting property of the impulse function →

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]$$

Q: How to represent $u[n]$ with unit impulses?

Discrete LTI Systems

- Consider a DT LTI system with input $x[n]$ and output $y[n]$
- *Define the **impulse response of the system*** to be the output when $x[n] = \delta[n]$ (i.e., the input is an impulse function).
  - Denote this impulse response by $h[n]$.



Discrete LTI Systems

- Consider a DT LTI system with input $x[n]$ and output $y[n]$
- System impulse response: system outputs $h[n]$ when input is $\delta[n]$
- Consider an arbitrary input signal

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]$$

- What is the LTI system output?
  - *Time-invariant:* at input $\delta[n-k]$, the system response is $h[n-k]$
  - *Linear (scaling property):* at input $x[k]\delta[n-k]$, the system response is $x[k]h[n-k]$
  - *Linear (additive property):* at input $x[n]$ (defined above), the system response is

Convolution sum $\rightarrow$ $\boldsymbol{y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]}$

Discrete LTI Systems

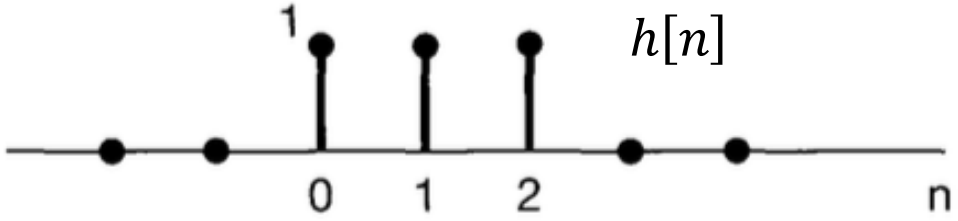
- The ***convolution*** of the signals $x[n]$ and $h[n]$ is denoted by

$$\boldsymbol{x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]}$$

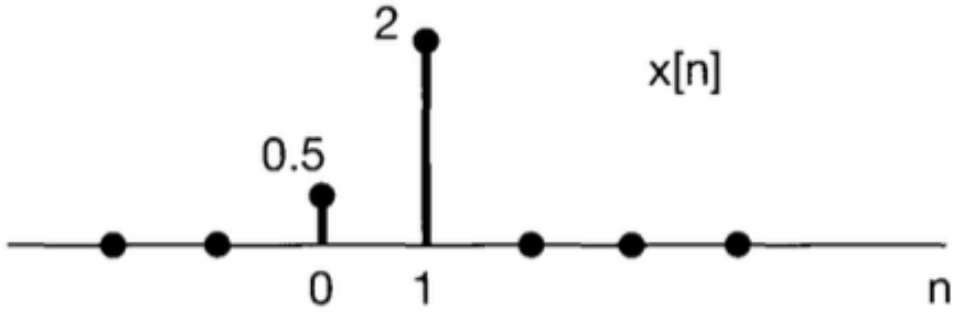
- For a **DT LTI system**: if $x[n]$ *is the input* to the system, and $h[n]$ *is the impulse response* of the system, then the *output of the system is* $y[n] = x[n] * h[n]$.
  - In other words, the output of any DT LTI system is a convolution of the input signal with the impulse response.
  - Any DT LTI system are completely characterized by its impulse response.

Discrete LTI Systems

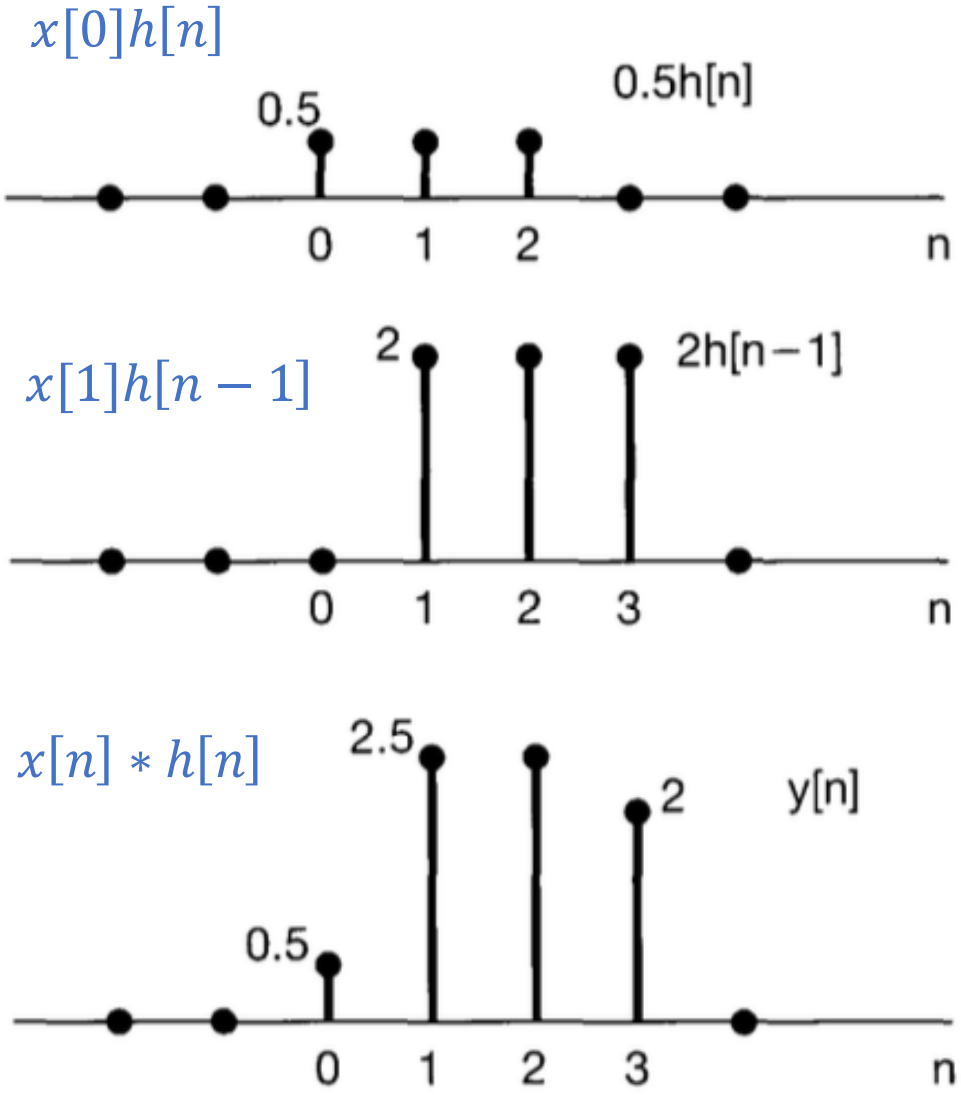
Q. Given the impulse of a DT LTI system $h[n]$:



If the input to this system is:

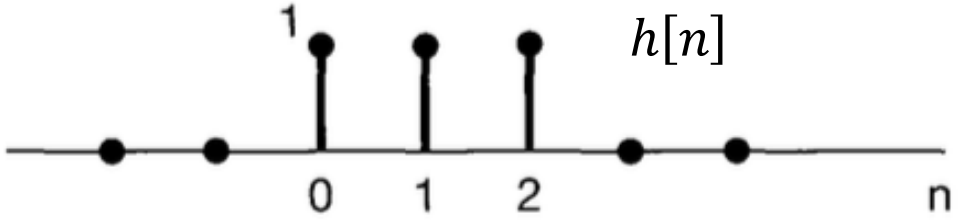


What is the response of the system?

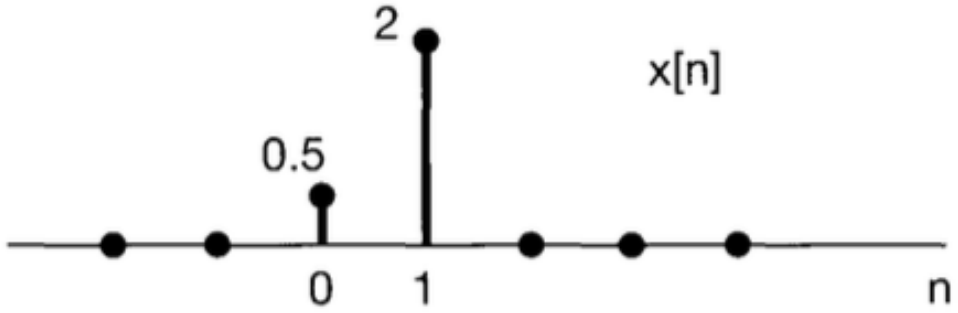


Discrete LTI Systems

Q. Given the impulse of a DT LTI system $h[n]$:



If the input to this system is:



What is the response of the system?

What if we are interested in the outp
value at a specific instance n?

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

$$\downarrow$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[-k+n]$$

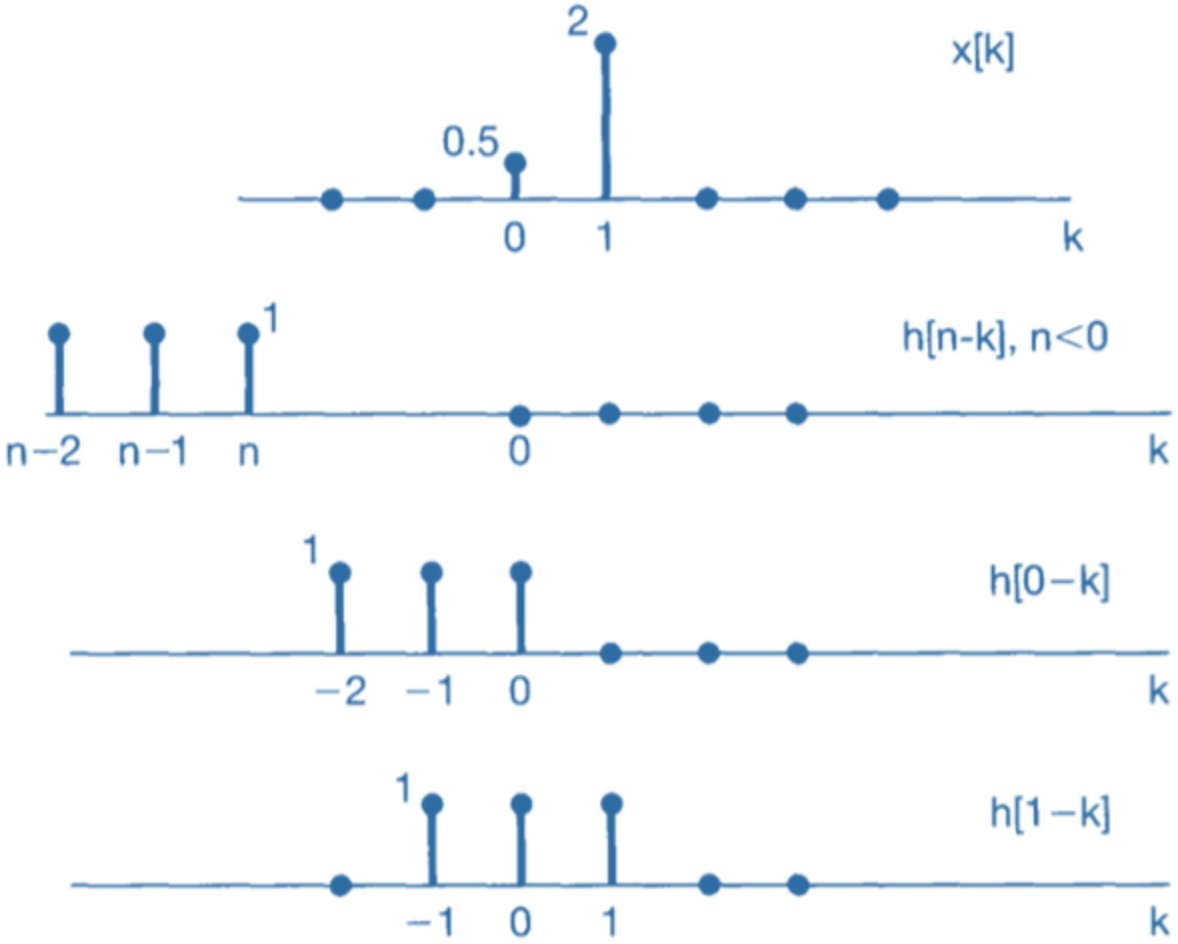
$x[k]$ & $h[-k+n]$ viewed as function of k

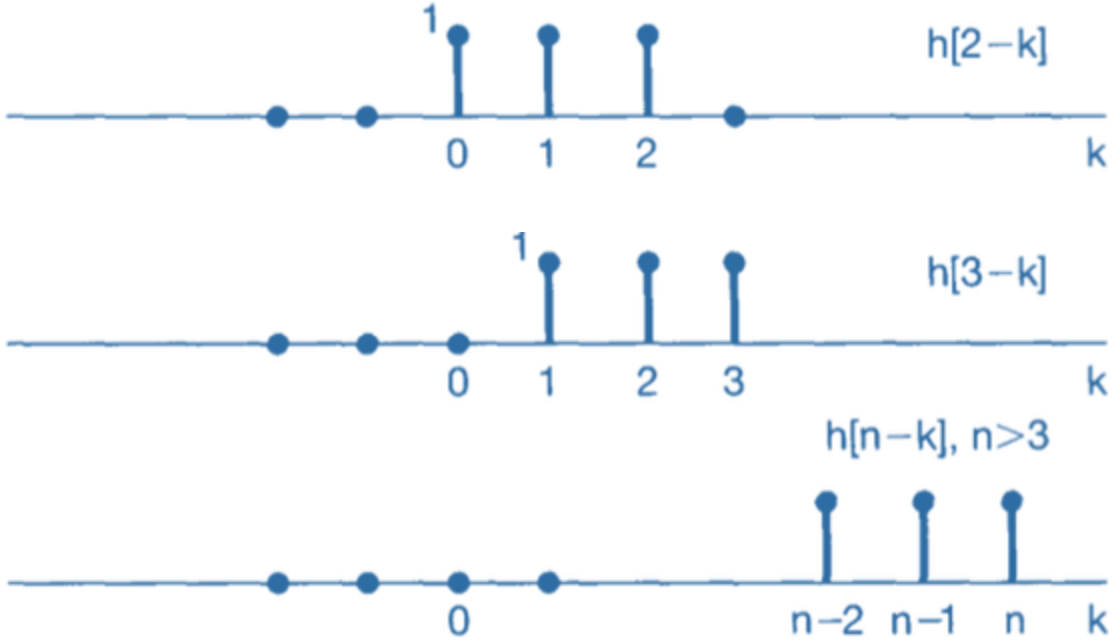




Discrete LTI Systems

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[-k+n]$$





$$y[0] = \sum_{k=-\infty}^{+\infty} x[k]h[0-k] = 0.5$$

$$y[1] = \sum_{k=-\infty}^{+\infty} x[k]h[1-k] = 0.5 + 2 = 2.5$$

$$y[2] = \sum_{k=-\infty}^{+\infty} x[k]h[2-k] = 0.5 + 2 = 2.5$$

$$y[3] = \sum_{k=-\infty}^{+\infty} x[k]h[3-k] = 2$$

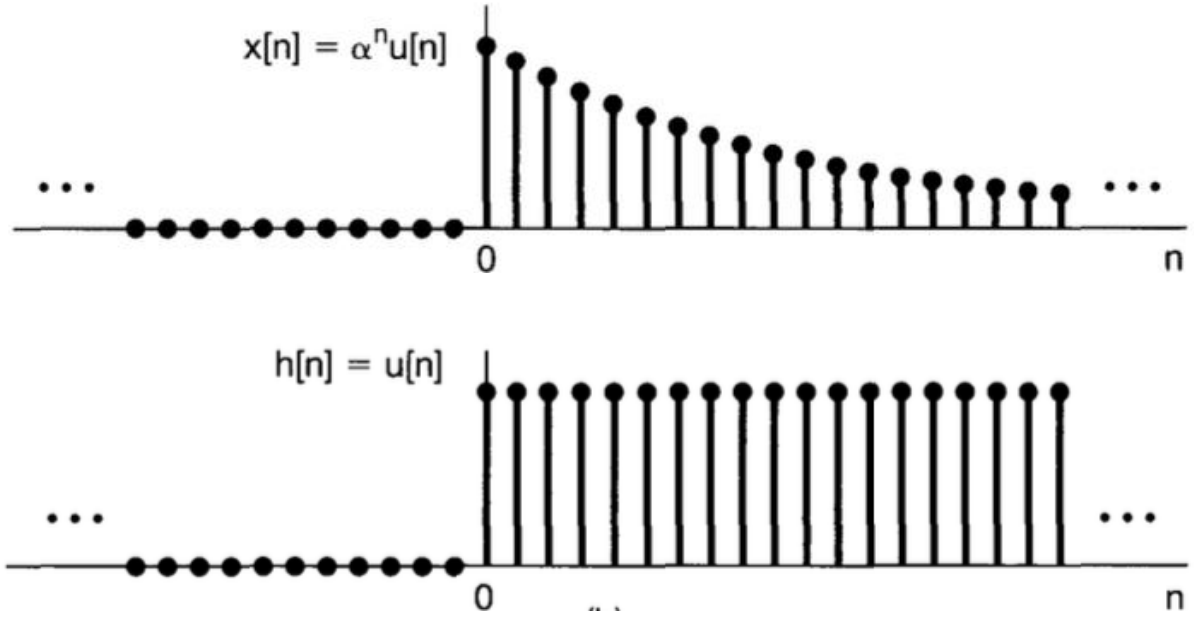
$n < 0$ or $n > 3$?

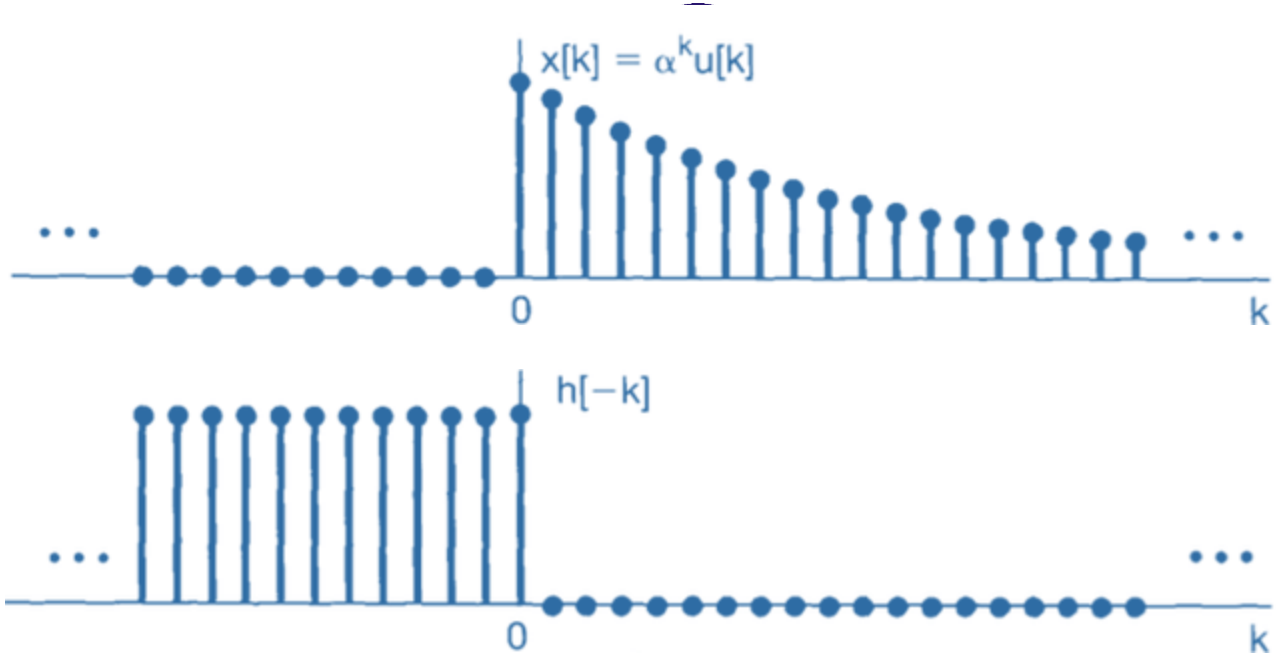
Discrete LTI Systems

Q. Input $x[n] = \alpha^n u[n]$, with $0 < \alpha < 1$,

Impulse response $h[n] = u[n]$,

Express output $y[n]$.





What intervals of n should we consider?

1) $n < 0$: $y = 0$

2) $n \geq 0$: $y = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}$

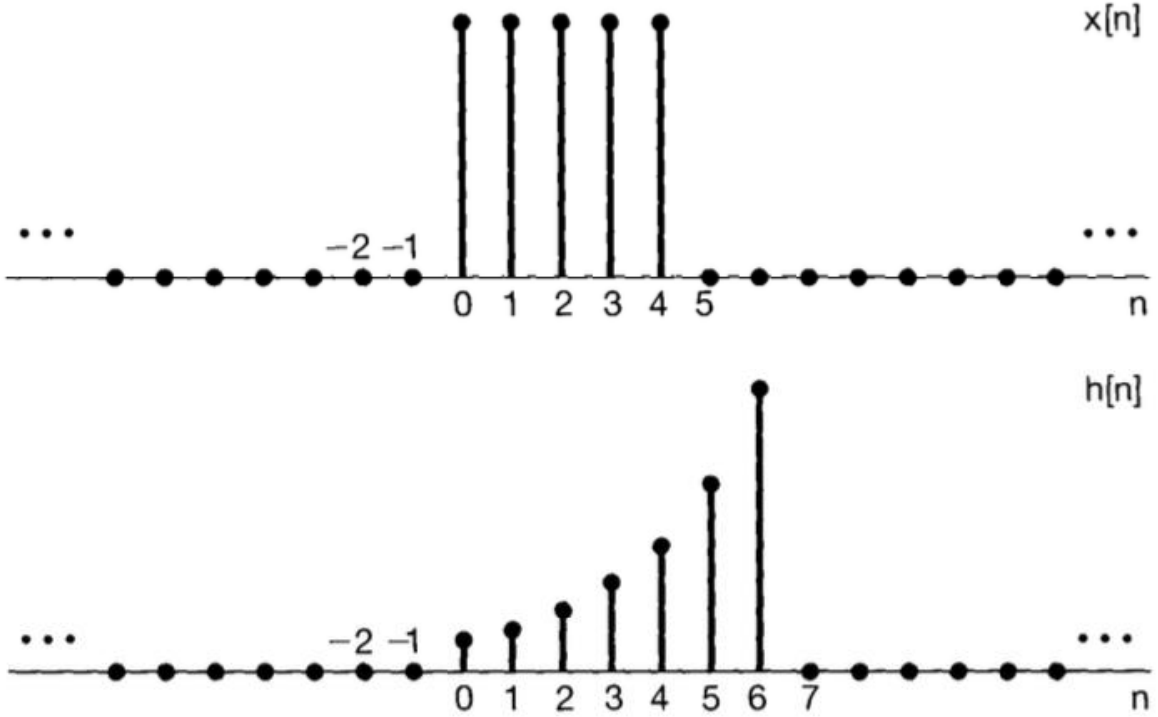
$$\Rightarrow y = \frac{1-\alpha^{n+1}}{1-\alpha} u[n]$$

Discrete LTI Systems

Q. Input $x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & otherwise \end{cases}$

Impulse response $h[n] = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & otherwise \end{cases}$

Express output $y[n]$.



What intervals of n should we consider?

1) $n < 0$
2) $0 \le n \le 4$
3) $5 \le n \le 6$
4) $7 \le n \le 10$
5) $n > 10$

→

$$y[n] = \begin{cases} 0, & n < 0 \\ \dfrac{1-\alpha^{n+1}}{1-\alpha}, & 0 \le n \le 4 \\ \dfrac{\alpha^{n-4}-\alpha^{n+1}}{1-\alpha}, & 4 < n \le 6 \\ \dfrac{\alpha^{n-4}-\alpha^{7}}{1-\alpha}, & 6 < n \le 10 \\ 0, & 10 < n \end{cases}$$

Self-practice:

Example 2.4, 2.5

Continuous LTI Systems

- For any signal $x(t)$, we can write:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

- For an LTI system where the *impulse response (when input is $\delta(t)$) is $h(t)$*, the output for a general input $x(t)$ is given by:

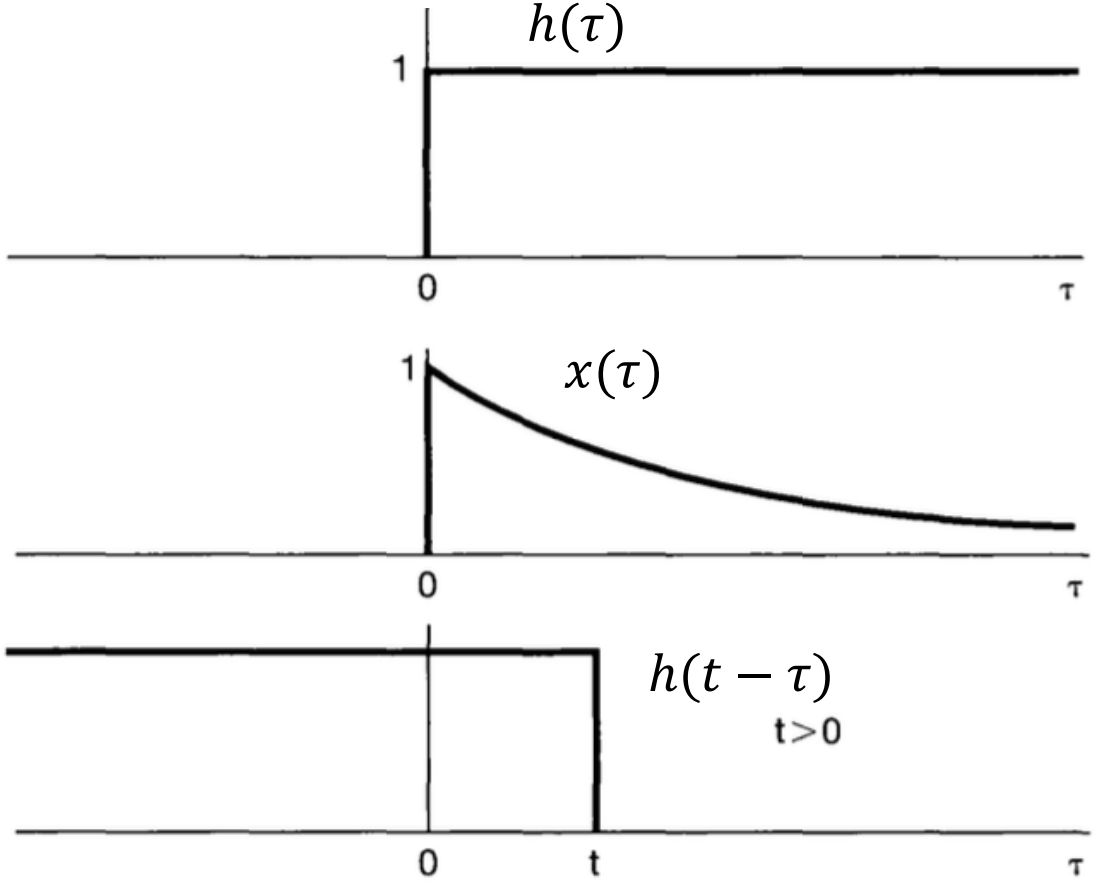
$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

- This is the **convolution integral**, denoted by $y(t) = x(t) * h(t)$

Continuous LTI Systems

Q. Consider an LTI system with unit impulse response $h(t) = u(t)$.

Let $x(t)$ be the input, where $x(t) = e^{-at}u(t)$, $a > 0$. Express the output $y(t)$.



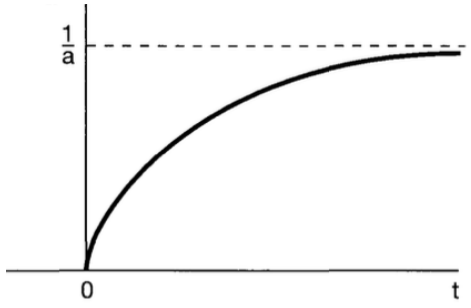
What intervals of t should we consider?

1) $t > 0$:

$$y(t) = \int_0^t e^{-a\tau} d\tau = \frac{1}{a}(1 - e^{-at})$$

2) otherwise: $y(t) = 0$

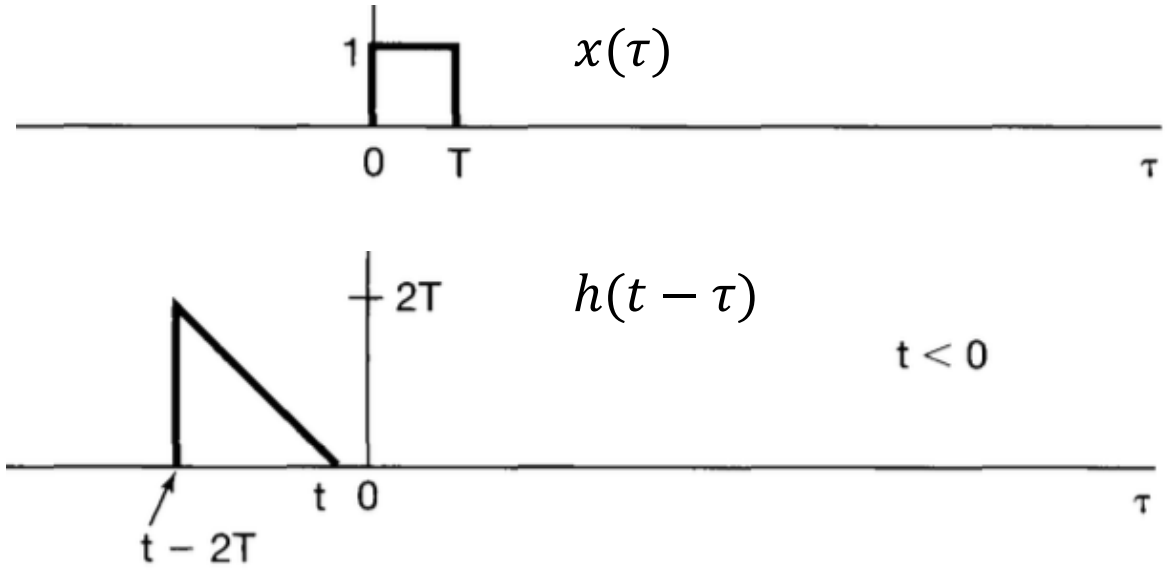
$$\Rightarrow y(t) = \frac{1}{a}(1 - e^{-at})u(t)$$



Continuous LTI Systems

Q. Consider the convolution of the following two signals:

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & otherwise \end{cases} \qquad h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & otherwise \end{cases}$$



What intervals of t should we consider?

1) $t < 0$
2) $0 < t < T$
3) $\mathrm{T} < t < 2T$
4) $2\mathrm{T} < t < 3T$
5) $t > 3T$

$$y(t) = \begin{cases} 0, \\ \frac{1}{2}t^2, \\ Tt - \frac{1}{2}T^2, \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2 \\ 0, \end{cases}$$

Self-practice: Example 2.7, 2.8

Properties of LTI Systems

- **Representation of CT & DT LTI systems** in terms of their unit impulse responses:
  - Convolution sum (DT LTI) & convolution integral (CT LTI)

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

→ Characteristics of an LTI system are *completely determined by its impulse response*

- This holds in general *only for LTI systems*

Properties of LTI Systems

1) The commutative property

$$x[n] * h[n] = h[n] * x[n]$$

$$x(t) * h(t) = h(t) * x(t)$$

- Proof?
- Let $r = n - k$, we have

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{r=-\infty}^{\infty} x[n-r]h[r] = h[n] * x[n]$$

- *The same holds for the continuous-time convolution*
- It *does not matter which of the signals we choose to flip & shift* in the convolution operation

Properties of LTI Systems

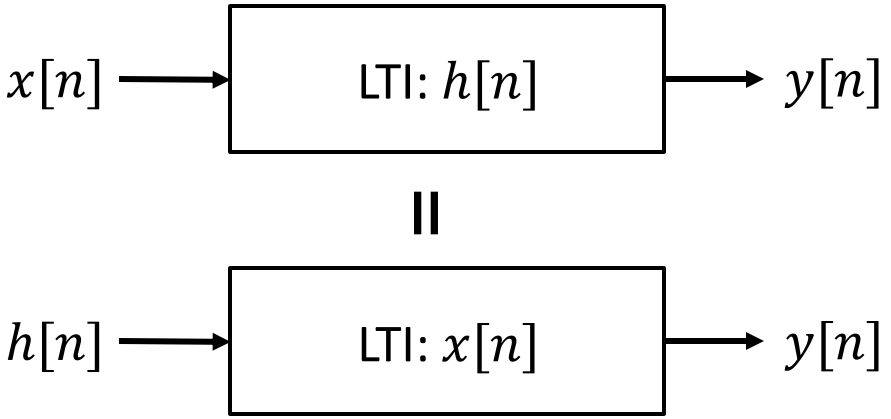
1) The commutative property

$$x[n] * h[n] = h[n] * x[n]$$

$$x(t) * h(t) = h(t) * x(t)$$

→ The output of an LTI system with input $x[n]$ & unit impulse response $h[n]$ is *identical* to the output of an LTI system with input $h[n]$ & unit impulse response $x[n]$

- *The same holds for CT LTI systems*



Properties of LTI Systems

2) The distributive property

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \big(h_1(t) + h_2(t)\big) = x(t) * h_1(t) + x(t) * h_2(t)$$

- Proof?

$$x[n] * (h_1[n] + h_2[n]) = \sum_{k=-\infty}^{\infty} x[k](h_1[n-k] + h_2[n-k])$$

$$= \sum_{k=-\infty}^{\infty} x[k]h_1[n-k] + \sum_{k=-\infty}^{\infty} x[k]h_2[n-k] = x[n] * h_1[n] + x[n] * h_2[n]$$

- *The same holds for the continuous-time convolution*

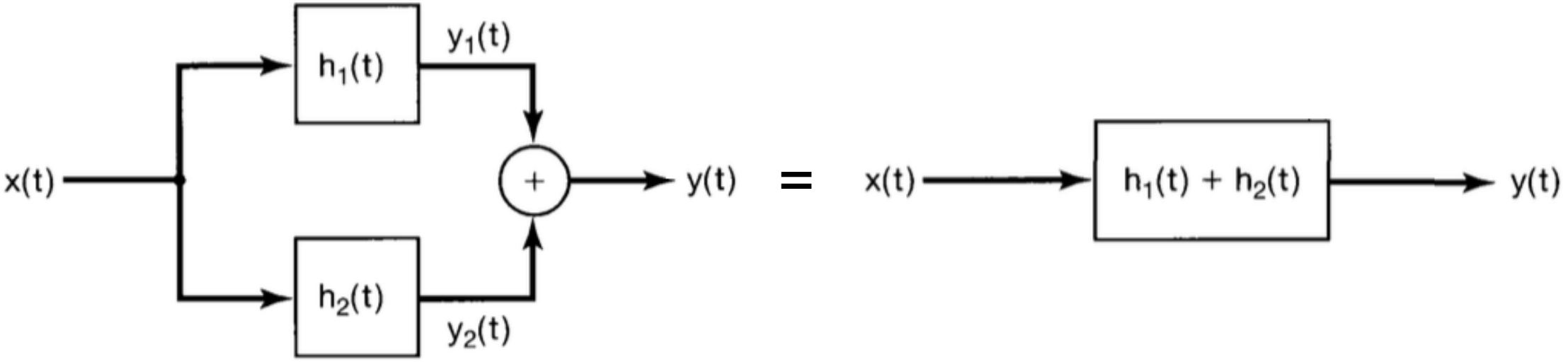
Properties of LTI Systems

2) The distributive property

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$x(t) * \big(h_1(t) + h_2(t)\big) = x(t) * h_1(t) + x(t) * h_2(t)$$

→ Consider two CT LTI systems *in parallel*,



- *The same holds for DT LTI systems*

Properties of LTI Systems

3) The associative property

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

$$x(t) * \big(h_1(t) * h_2(t)\big) = \big(x(t) * h_1(t)\big) * h_2(t)$$

- Proof? *(The same holds for the continuous-time convolution)*

$$x[n] * (h_1[n] * h_2[n]) = \sum_{k_1=-\infty}^{\infty} x[k_1] \left(\sum_{k_2=-\infty}^{\infty} h_1[k_2] h_2[(n-k_1)-k_2] \right)$$

$$= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x[k_1]\, h_1[(k_1+k_2)-k_1] h_2[n-(k_1+k_2)]$$

$$= \sum_{k_3=-\infty}^{\infty} \sum_{k_1=-\infty}^{\infty} x[k_1]\, h_1[k_3-k_1] h_2[n-k_3] = (x[n] * h_1[n]) * h_2[n]$$

- It *does not matter which order we do the convolutions*

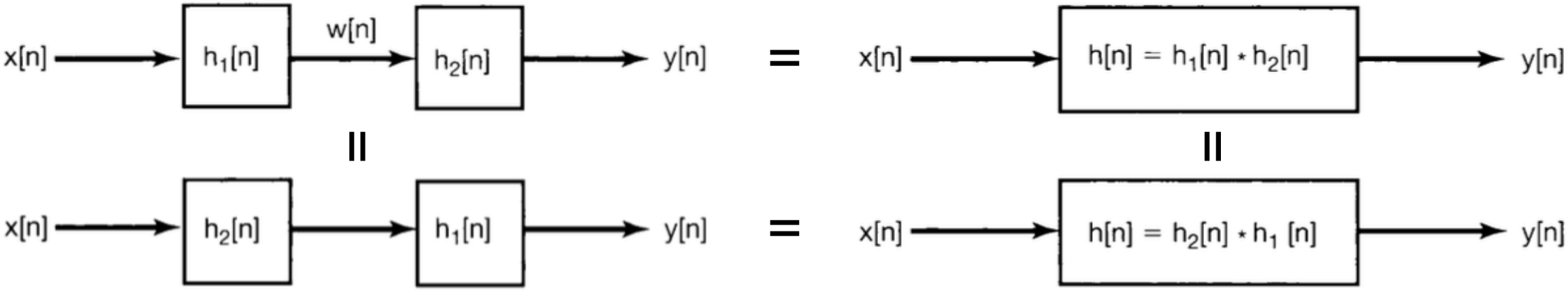
Properties of LTI Systems

3) The associative property

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

$$x(t) * \big(h_1(t) * h_2(t)\big) = \big(x(t) * h_1(t)\big) * h_2(t)$$

→ Consider LTI systems *in series (in cascade)*, the overall system response *does not depend on the order of the systems in series (in cascade).*



- *The same holds for CT LTI systems*

Properties of LTI Systems

4) Memoryless or with memory

- Recall that a system is *memoryless* if its output at any time *depends on the value of the input at that time*
- For **LTI systems**, let $h[n]$ (or $h(t)$) be the impulse response of a given LTI system, we have

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} x[n-k]h[k]$$

- For an LTI system to be memoryless, we require that $h[k] = 0$ for all $k \neq 0$. In other words, $\boldsymbol{h[k] = K\delta[n]}$ **for some constant** $\boldsymbol{K}$.
- Why?
  - Otherwise, $y[n]$ will depend on a value of the input signal other than at time-step n.

Properties of LTI Systems

4) Memoryless or with memory

- For **LTI systems**, let $h[n]$ (or $h(t)$) be the impulse response of a given LTI system, we have

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} x[n-k]h[k]$$

- A DT LTI system is **memoryless iff $h[n] = K\delta[n]$ for some constant K**.
- A CT LTI system is **memoryless iff $h(t) = K\delta(t)$ for some constant K**.

→ All **LTI memoryless systems** have the form

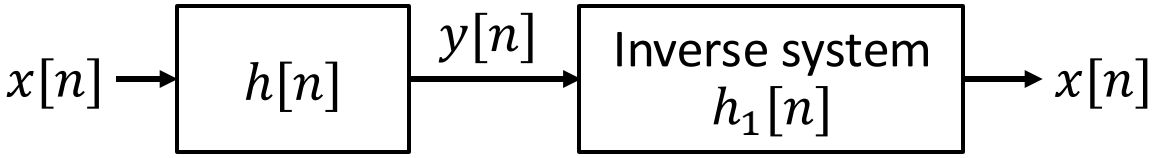
$$y[n] = Kx[n] \qquad \text{or} \qquad y(t) = Kx(t)$$

for some constant K.

Properties of LTI Systems

5) Invertibility of LTI systems

- Recall that a system is *invertible* if the output of the system uniquely specifies the input.
- Consider an LTI system with impulse response $h[n]$. Suppose it has an inverse system with impulse response $h_1[n]$.



$$x[n] = (x[n] * h[n]) * h_1[n]$$

- By associative property, $x[n] = x[n] * (h[n] * h_1[n])$ for all input signals $x[n]$, thus we have $\boldsymbol{h[n] * h_1[n] = \delta[n]}$.
- Similarly, the two LTI systems are inverses of each other if $\boldsymbol{h(t) * h_1(t) = \delta(t)}$

Properties of LTI Systems

Q. Consider the LTI system consisting of a pure time shift

$$y(t) = x(t - t_0)$$

- The impulse response? $h(t) = \delta(t - t_0)$
- For a general signal $x(t)$, the output: $y(t) = x(t - t_0) = x(t) * \delta(t - t_0)$

$$\boldsymbol{x(t - t_0) = x(t) * \delta(t - t_0)}$$

- The convolution of a signal with a shifted impulse *shifts* the signal.

- The inverse system? $h_1(t) = \delta(t + t_0)$
- Verification: $h(t) * h_1(t) = \delta(t - t_0) * \delta(t + t_0) = \delta(t)$

- The same applies to the DT LTI system.

Properties of LTI Systems

Q. Consider the LTI system with impulse response: $h[n] = u[n]$

- For a general signal $x[n]$, the output:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]u[n-k] = \sum_{k=-\infty}^{n} x[k]$$

- Inverse system? $x[n] - x[n-1]$

- Verification:
  - Impulse response of the inverse system: $h_1[n] = \delta[n] - \delta[n-1]$

$$h[n] * h_1[n] = u[n] * (\delta[n] - \delta[n-1]) = u[n] * \delta[n] - u[n] * \delta[n-1] = \delta[n]$$

Properties of LTI Systems

6) Causality of LTI systems

- Recall that a system is *causal* if its output at time t *depends only on the inputs up to (including)* t.
- For **LTI systems**, let $h[n]$ (or $h(t)$) be the impulse response, we have

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} x[n-k]h[k]$$

- A DT LTI system is **causal iff $h[n]$ is 0 for all $n < 0$**.
- A CT LTI system is **causal iff $h(t)$ is 0 for all $t < 0$**.
- For a causal LTI system, the above condition implies:

$$y[n] = \sum_{k=0}^{+\infty} h[k]x[n-k] = \sum_{k=-\infty}^{n} x[k]h[n-k] \qquad y(t) = \int_{0}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)h(t-\tau)d\tau$$

Properties of LTI Systems

7) Stability of LTI systems

- Recall that a system is *stable* if *every bounded input produces a bounded output*.
- Consider an LTI system: $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$, we have

$$|y[n]| = \left| \sum_{k=-\infty}^{+\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{+\infty} |h[k]x[n-k]| = \sum_{k=-\infty}^{+\infty} |h[k]||x[n-k]|$$

- Suppose that $x[n]$ is bounded, *i.e.*, $|x[n-k]| \leq B$ for all values of k and n, this implies

$$|y[n]| \leq B \sum_{k=-\infty}^{+\infty} |h[k]|$$

- The system is ***stable*** (i.e., $y[n]$ is bounded in magnitude) *iff the impulse response is absolutely summable*, that is

$$\sum_{\boldsymbol{k=-\infty}}^{+\infty} |\boldsymbol{h[k]}| < \infty$$

Properties of LTI Systems

7) Stability of LTI systems

- Recall that a system is *stable* if *every bounded input produces a bounded output*.
- How about a CT LTI system?

$$|y(t)| = \left| \int_{-\infty}^{+\infty} h(\tau)\, x(t-\tau) d\tau \right| \leq \int_{-\infty}^{+\infty} |h(\tau)|\, |x(t-\tau)| d\tau$$

- Suppose that $x(t)$ is bounded, *i.e.*, $|x(t-\tau)| \leq B$ for all values of t and τ, this implies

$$|y(t)| \leq B \int_{-\infty}^{+\infty} |h(\tau)|\, d\tau$$

- The system is ***stable*** (i.e., $y(t)$ is bounded in magnitude) *iff the impulse response is absolutely integrable*, that is

$$\int_{-\infty}^{+\infty} |\boldsymbol{h(\tau)}|\, \boldsymbol{d\tau} < \infty$$

Properties of LTI Systems

8) Step Response of LTI systems

- **Unit step response: $s[n]$ or $s(t)$**
  - Output when $x[n] = u[n]$, *or* $x(t) = u(t)$
  - A signal that is also used often in *describing the behavior of LTI systems*
- **Relationship with the impulse response**
  - The step response of a DT LTI system is the convolution of the unit step ($u[n]$) with the impulse response ($h[n]$). That is,

$$\boldsymbol{s[n] = u[n] * h[n] = \sum_{k=-\infty}^{\infty} u[n-k]h[k] = \sum_{k=-\infty}^{n} h[k]}$$

  - This is equivalent to $s[n] = s[n-1] + h[n]$; or $\boldsymbol{h[n] = s[n] - s[n-1]}$
  - Thus, the step response of a DT LTI system is the running sum of the impulse responses.

Properties of LTI Systems

8) Step Response of LTI systems

- **Unit step response: $s[n]$ or $s(t)$**
  - Output when $x[n] = u[n]$, *or* $x(t) = u(t)$
  - A signal that is also used often in *describing the behavior of LTI systems*
- **Relationship with the impulse response**
  - How about a CT LTI system?

$$s(t) = \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} u(t-\tau)h(\tau)d\tau = \int_{-\infty}^{t} h(\tau)d\tau$$

$$\frac{ds(t)}{dt} = h(t)$$

  - Thus, the step response of a CT LTI system is the running integral of the impulse responses. The impulse response is the derivative of the step response.

Summary

- **DT & CT LTI Systems**
  - Representation in terms of their unit impulse responses
  - Convolution sum (DT LTI) & convolution integral (CT LTI)
  - Convolution calculations: by formula, by graphical approach
- **Properties of LTI Systems**
  - Commutative; distributive; associative: convolution property & what each property means for LTI systems
  - Memoryless or with memory: relationship with impulse response
  - Invertibility: relationship with impulse response; find inverse system
  - Causality: relationship with impulse response; causal system representation
  - Stability: relationship with impulse response
  - Step response: relationship with impulse response